

Unsupervised Learning Analysis of Wine Data

Using Principal Component Analysis, K-Means Clustering, and Hierarchical Clustering

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Introduction

Unsupervised learning methods are essential for identifying structure in unlabeled data. Principal Component Analysis (PCA) reduces dimensionality by transforming correlated variables into orthogonal components that retain most of the dataset's variance (Jolliffe & Cadima, 2016). Clustering techniques such as K-Means and Hierarchical Clustering then group observations based on similarity, revealing natural patterns within the data (Xu & Tian, 2015).

This assignment required completing Conceptual Question #1 and Applied Question #9 from *ISLR Python*, conducting PCA to retain at least 80% variance, evaluating multiple K-Means cluster solutions, performing hierarchical clustering, and interpreting all findings. The Kaggle wine dataset includes 178 observations and 13 chemical attributes describing wine composition. The goal of this analysis was to reduce dimensionality, investigate assumptions, and identify meaningful clusters within the wine data.

Conceptual Question #1 (ISLR Python, p. 552)

Why is scaling important before applying PCA?

Principal Component Analysis (PCA) identifies directions of maximum variance in a dataset, but the method is sensitive to the scale of the variables. When predictors are measured in different units (e.g., Alcohol in percent vs. Proline in mg/L), variables with larger numerical ranges dominate the variance structure and disproportionately influence the principal components. Standardizing all variables to have mean 0 and variance 1 ensures that PCA reflects true correlation-based structure rather than raw magnitude differences. Without scaling, PCA

would incorrectly emphasize high-variance variables, leading to misleading component loadings and distorted interpretations (James et al., 2023).

Applied Question #9 (ISLR Python, p. 554)

Perform K-Means clustering on a dataset and interpret the results.

K-Means clustering partitions observations into k groups by minimizing within-cluster variation. After standardizing the predictors, multiple values of k were evaluated using the Elbow Method and Silhouette Analysis. The Elbow plot showed a sharp decrease in inertia from $k = 2$ to $k = 3$, followed by diminishing returns. Silhouette scores confirmed that $k = 3$ produced the strongest cohesion and separation among clusters.

Cluster profiling revealed meaningful differences in chemical composition. One cluster exhibited higher Alcohol, Total Phenols, and Flavanoids, suggesting richer phenolic content. Another cluster showed higher Malic Acid and lower Magnesium, while the third cluster contained wines with lower Alcohol and moderate phenolic levels. These results demonstrate how K-Means can uncover natural groupings in multivariate data and support interpretation of underlying chemical patterns (Xu & Tian, 2015).

Exploratory Data Analysis and Assumption Investigation

Initial exploratory data analysis confirmed that the dataset contained no missing values and no duplicate observations, making it suitable for PCA and clustering. Descriptive statistics showed substantial variation across variables such as Alcohol, Malic Acid, Color Intensity, and Proline.

Correlation analysis revealed strong relationships among phenolic compounds, including Flavanoids, Total Phenols, OD280, and Proline. The correlation matrix showed values exceeding

0.70 among these variables, indicating redundancy and supporting PCA as an appropriate dimensionality reduction technique.

Feature distribution plots showed mild skewness in variables such as Color Intensity and Proline, but no extreme outliers. Because PCA and distance-based clustering are sensitive to scale, all variables were standardized using z-score normalization (James et al., 2023). This ensured equal weighting across features during PCA and clustering.

Principal Component Analysis

PCA was applied to the standardized dataset. The first principal component explained 36.20% of the variance, and the second explained 19.21%, meaning the first two components captured over half of the dataset's variability.

A cumulative variance plot demonstrated that five principal components were required to retain at least 80% of the total variance, satisfying the assignment requirement. The reduced dataset shape was (178, 5) compared to the original (178, 13).

The PCA loading matrix revealed that Flavanoids, Total Phenols, OD280, Proline, and Proanthocyanins contributed most strongly to PC1. PC2 was influenced by Alcohol and Color Intensity, while PC3 reflected ash-related variables (Ash and Ash Alcanity). These results indicate that phenolic content and alcohol concentration are major drivers of variation among wines.

K-Means Clustering

K-Means clustering was performed using the PCA-reduced dataset. Values of k from 2 to 10 were evaluated using the Elbow Method. Inertia decreased sharply from $k = 2$ to $k = 3$, then leveled off, suggesting diminishing returns beyond three clusters.

Silhouette Analysis confirmed $k = 3$ as the optimal solution, producing the highest silhouette score (0.3691). This indicates strong cohesion within clusters and clear separation between them (Xu & Tian, 2015).

Cluster membership counts were: Cluster 0: 51 observations, Cluster 1: 62 observations, and Cluster 2: 65 observations. Cluster profiling revealed meaningful differences. Cluster 1 exhibited the highest Alcohol (13.68), Magnesium (108), Total Phenols (2.85), and Flavanoids (3.00), suggesting wines with richer chemical composition. Cluster 0 had the highest Malic Acid (3.31), while Cluster 2 showed lower Alcohol (12.25) and Magnesium (92.74). Visualization of clusters in PCA space showed clear separation, confirming that K-Means effectively distinguished distinct wine groups.

Hierarchical Clustering

Hierarchical Clustering was conducted using complete linkage and Euclidean distance. The dendrogram revealed three major branches, consistent with the K-Means solution. Cutting the dendrogram at three clusters produced groups of: Cluster 1: 23 observations, Cluster 2: 48 observations, and Cluster 3: 107 observations

The cophenetic correlation coefficient was 0.5433, indicating moderate fidelity between the dendrogram and actual pairwise distances. This suggests that hierarchical clustering captured meaningful structure but was less precise than K-Means in representing distance relationships.

Comparison of Clustering Methods

A cross-tabulation comparing K-Means and hierarchical cluster assignments showed partial alignment. One K-Means cluster (Cluster 1) mapped entirely to Hierarchical Cluster 3 (62 observations). Other clusters showed mixed alignment, reflecting differences in algorithmic assumptions.

K-Means produced more balanced clusters and stronger validation metrics, while hierarchical clustering provided a broader structural view through its dendrogram. These differences are consistent with prior research showing that clustering algorithms may reveal different aspects of underlying data structure (Xu & Tian, 2015).

Conclusions

This analysis successfully applied PCA, K-Means clustering, and hierarchical clustering to the wine dataset. PCA reduced the dataset from 13 variables to five principal components while retaining 80.16% of the variance. K-Means clustering identified three distinct clusters, supported by the highest silhouette score and clear separation in PCA space. Hierarchical clustering also revealed three clusters, though with different membership distributions and moderate dendrogram reliability.

Overall, PCA proved effective for dimensionality reduction, and both clustering methods identified meaningful patterns in wine composition. These unsupervised learning techniques provide valuable tools for exploring complex datasets and uncovering natural groupings within data (James et al., 2023; Jolliffe & Cadima, 2016).

References

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